

Q85 - Subset Generation in Lexicographic Order

Aim: Generate all subsets of a given set in lexicographic order using Backtracking.

Input: Ordered set of elements.

Output: All subsets in lexicographic sequence.

Algorithm:

1. Sort the input elements.
2. Print the current subset.
3. Recursively choose subsequent elements in sorted order.
4. Backtrack by removing the last chosen element.

Backtracking restores the subset state after each recursive call, ensuring correctness.

Time Complexity: $O(2^n)$

Result: All subsets were generated exactly once in lexicographic order.

Conclusion: Backtracking systematically generates subsets while preserving lexicographic ordering.

Output Screenshot: Insert terminal output screenshot here.